

## CLINTON, AR, USA

WMO: 720401

Lat: 35.600N

Lon: 92.450W

Elev: 157

StdP: 99.45

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00133

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | -9.1       | -7.1 | -17.2                         | 0.8 | -5.5 | -13.9 | 1.1 | -3.2 | 8.0                   | 8.1  | 7.3 | 6.9  | 0.8                      | 320  | 0.373 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.2                   | 37.0            | 24.8 | 34.9 | 25.1 | 33.0 | 24.8 | 27.1                | 32.3 | 26.4 | 31.9 | 25.8 | 31.3 | 1.9                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 26.0                            | 21.8 | 29.9 | 25.0 | 20.5 | 29.1 | 24.1 | 19.3 | 28.2 | 86.4          | 32.5 | 82.8 | 32.0 | 80.7 | 31.4 | 31.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 6.6           | 5.4  | 4.6 | DB                         | N/A   | N/A                | N/A | N/A                                                | N/A   | N/A        | N/A   | N/A        | N/A   | N/A        | N/A   | N/A  | N/A |
| (5)               |      |     | WB                         | -13.0 | 28.6               | 3.3 | 1.3                                                | -15.4 | 29.5       | -17.4 | 30.2       | -19.3 | 30.9       | -21.7 | 31.8 |     |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 15.5   | 2.7   | 6.0   | 10.4  | 15.3  | 20.1  | 25.2  | 26.5  | 26.1  | 22.6  | 16.0  | 9.7   | 5.1   |
|                                                                        | DBStd       | 9.29   | 5.21  | 5.57  | 5.00  | 4.04  | 3.85  | 2.48  | 2.56  | 2.69  | 3.36  | 4.76  | 4.65  | 4.82  |
|                                                                        | HDD10.0     | 670    | 235   | 137   | 56    | 7     | 1     | 0     | 0     | 0     | 0     | 7     | 61    | 167   |
|                                                                        | HDD18.3     | 1997   | 486   | 347   | 248   | 105   | 29    | 0     | 0     | 0     | 5     | 104   | 261   | 412   |
|                                                                        | CDD10.0     | 2679   | 8     | 24    | 68    | 166   | 313   | 457   | 511   | 499   | 377   | 191   | 52    | 15    |
|                                                                        | CDD18.3     | 965    | 0     | 0     | 2     | 14    | 83    | 207   | 252   | 241   | 132   | 31    | 2     | 1     |
|                                                                        | CDH23.3     | 9926   | 0     | 7     | 57    | 207   | 776   | 2168  | 2696  | 2431  | 1223  | 323   | 22    | 16    |
| (13)                                                                   | CDH26.7     | 4567   | 0     | 0     | 5     | 28    | 236   | 1042  | 1397  | 1234  | 529   | 82    | 2     | 11    |
| (14)                                                                   | Wind        | WSAvg  | 1.0   | 1.2   | 1.4   | 1.5   | 1.6   | 1.1   | 0.9   | 0.7   | 0.6   | 0.9   | 1.0   | 1.1   |
| (15) Precipitation                                                     | PrecAvg     | 1331   | 104   | 92    | 125   | 139   | 133   | 98    | 94    | 78    | 110   | 121   | 118   | 121   |
|                                                                        | PrecMax     | 1893   | 189   | 200   | 321   | 295   | 282   | 166   | 186   | 200   | 243   | 379   | 304   | 188   |
|                                                                        | PrecMin     | 926    | 23    | 30    | 48    | 34    | 28    | 50    | 37    | 17    | 1     | 0     | 37    | 6     |
|                                                                        | PrecStd     | 264    | 46    | 48    | 68    | 58    | 67    | 32    | 38    | 45    | 60    | 91    | 69    | 53    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 20.2  | 24.8  | 27.3  | 28.9  | 32.4  | 38.6  | 38.9  | 38.7  | 35.9  | 31.1  | 26.2  | 22.4  |
|                                                                        |             | MCWB   | 16.2  | 17.7  | 18.2  | 21.2  | 22.5  | 22.6  | 23.8  | 25.5  | 24.6  | 22.3  | 18.8  | 18.3  |
|                                                                        | 2%          | DB     | 17.7  | 21.1  | 25.0  | 27.4  | 31.0  | 34.9  | 37.0  | 36.4  | 33.0  | 28.9  | 22.8  | 18.8  |
|                                                                        |             | MCWB   | 13.4  | 15.4  | 16.8  | 18.4  | 22.5  | 24.8  | 24.5  | 25.6  | 24.2  | 22.3  | 16.2  | 16.0  |
|                                                                        | 5%          | DB     | 14.7  | 18.7  | 22.5  | 26.0  | 29.0  | 33.0  | 34.8  | 34.1  | 32.2  | 27.3  | 21.0  | 16.2  |
|                                                                        |             | MCWB   | 10.2  | 14.3  | 15.5  | 17.8  | 22.2  | 24.6  | 25.4  | 24.6  | 24.2  | 19.8  | 16.3  | 12.1  |
|                                                                        | 10%         | DB     | 12.3  | 16.3  | 20.0  | 23.9  | 27.6  | 32.2  | 32.9  | 32.7  | 30.0  | 24.9  | 18.6  | 12.9  |
|                                                                        |             | MCWB   | 8.6   | 12.4  | 14.3  | 16.9  | 21.3  | 24.5  | 25.4  | 24.7  | 22.9  | 18.7  | 14.7  | 10.2  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 17.7  | 18.8  | 19.8  | 22.2  | 25.0  | 27.2  | 28.3  | 28.1  | 26.2  | 24.5  | 20.7  | 19.3  |
|                                                                        |             | MCDB   | 19.2  | 22.1  | 25.8  | 27.0  | 29.6  | 32.7  | 33.0  | 33.5  | 32.4  | 28.9  | 23.3  | 21.2  |
|                                                                        | 2%          | WB     | 15.2  | 17.3  | 18.6  | 20.7  | 24.0  | 26.4  | 27.2  | 27.1  | 25.5  | 23.3  | 18.8  | 16.8  |
|                                                                        |             | MCDB   | 17.1  | 19.2  | 23.6  | 25.0  | 28.4  | 31.9  | 32.0  | 32.6  | 31.2  | 27.5  | 21.5  | 18.2  |
|                                                                        | 5%          | WB     | 11.2  | 15.2  | 17.0  | 19.8  | 23.2  | 25.7  | 26.4  | 26.3  | 24.8  | 22.0  | 16.9  | 13.3  |
|                                                                        |             | MCDB   | 13.4  | 17.8  | 20.5  | 23.5  | 27.4  | 31.2  | 31.7  | 32.0  | 30.0  | 25.0  | 19.4  | 15.1  |
|                                                                        | 10%         | WB     | 8.0   | 12.7  | 15.4  | 18.8  | 22.4  | 25.1  | 25.7  | 25.6  | 24.0  | 20.3  | 15.1  | 11.3  |
|                                                                        |             | MCDB   | 11.1  | 15.3  | 18.7  | 22.0  | 26.4  | 30.6  | 31.1  | 31.1  | 28.5  | 23.0  | 18.0  | 12.7  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 12.1  | 12.2  | 13.4  | 14.0  | 12.2  | 12.3  | 12.2  | 12.2  | 12.8  | 14.2  | 13.4  | 10.9  |
|                                                                        |             | MCDBR  | 16.5  | 16.1  | 17.5  | 16.5  | 14.4  | 14.3  | 14.3  | 14.8  | 14.7  | 15.4  | 16.2  | 15.1  |
|                                                                        |             | MCWBR  | 11.3  | 11.3  | 10.2  | 8.7   | 7.2   | 5.7   | 4.8   | 5.4   | 6.3   | 7.8   | 10.4  | 11.3  |
|                                                                        | 5% WB       | MCDBR  | 12.7  | 13.0  | 14.8  | 12.4  | 11.6  | 12.1  | 11.9  | 12.6  | 12.7  | 12.1  | 13.4  | 12.8  |
|                                                                        |             | MCWBR  | 10.3  | 10.1  | 9.7   | 7.9   | 6.4   | 5.6   | 5.2   | 5.2   | 5.8   | 6.9   | 10.0  | 11.1  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.317 | 0.322 | 0.357 | 0.391 | 0.426 | 0.451 | 0.483 | 0.477 | 0.405 | 0.356 | 0.328 | 0.307 |
|                                                                        | taud        |        | 2.484 | 2.467 | 2.403 | 2.309 | 2.270 | 2.214 | 2.176 | 2.168 | 2.357 | 2.475 | 2.504 | 2.526 |
|                                                                        | Ebn at Noon |        | 884   | 922   | 912   | 893   | 862   | 837   | 807   | 804   | 851   | 871   | 864   | 875   |
|                                                                        | Edh at Noon |        | 84    | 95    | 111   | 128   | 135   | 143   | 148   | 146   | 115   | 94    | 82    | 76    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.44  | 2.97  | 4.08  | 5.13  | 5.59  | 6.38  | 6.14  | 5.68  | 4.85  | 3.71  | 2.64  | 2.13  |
|                                                                        | RadStd      |        | 0.21  | 0.38  | 0.29  | 0.37  | 0.46  | 0.48  | 0.38  | 0.37  | 0.38  | 0.48  | 0.29  | 0.24  |

## Historical Trends

|                              | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (26 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page